



Anuj Kumar

Master of Engineering
in Mechanical Engineering
Chandigarh University, Gharuan, Mohali

+91-6287270392

anuj1009kumar@gmail.com
www.linkedin.com/in/anuj1009

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Master of Engineering	Chandigarh University, Gharuan, Mohali	6.9 (Till 3 rd sem)	2024-2026
Bachelor of Engineering	P.U. Swami Sarvanand Giri Regional Centre, Bajwara, Hoshiarpur	7.8	2020-2023
Diploma	GGs Polytechnic, Kharar, Mohali	73.3%	2017-2020
Secondary	Bihar School Examination Board	74%	2017

Master of Engineering – RESEARCH

Investigation of Milling Process Parameters for Optimization of Surface Roughness and Material Removal Rate in Al-6061 Aluminium Alloy

June 2025 – Dec. 2025

Conference: CHARM-2025, Guidance of Dr. Kamaljeet Singh

- MRR** increased by 16.5% under optimized cutting parameters (V1F1D3) when compared with the initial settings (V1F1D1). The depth of cut was the most dominant factor influencing MRR. The optimal MRR condition was achieved at cutting speed (V) = 350 rev/min, feed rate (F) = 120 mm/min and depth of cut (D) = 0.60 mm (V1F1D3).
- SR** was reduced by 30.6% under optimized cutting parameters (V3F1D1) when compared to the initial trial (V1F1D1). Cutting speed, feed rate, and depth of cut jointly influenced SR, with cutting speed showing a strong effect. The optimal SR condition was obtained at cutting speed (V) = 700 rev/min, feed rate (F) = 120 mm/min and depth of cut (D) = 0.20 mm (V3F1D1).

Optimization of Material Removal Rate (MRR) in Dry Turning of EN8 Steel Using CNC Lathe: A Taguchi-Based Approach

Jan

2025 - Apr 2025

Conference: ICAMSF-2025, Guidance of Dr. Kamaljeet Singh

- The highest contribution (40.94%) to MRR, following by feed rate (36.43%) and spindle speed (20.14%). Machining parameters are A1B1C3, i.e., spindle speed = 550 m/min., feed rate = 0.14 mm/rev. and depth of cut = 1.2 mm.

Bachelor of Engineering - PROJECTS

Vehicle Speed Limiter

June 2021 - Dec

2021 Course Project, Under Guidance of Dr. Gaurav Saini

- A **vehicle speed limiter** (or speed governor) is an electronic-mechanical device that restricts a vehicle's maximum speed by controlling engine fuel flow or power, preventing the driver from exceeding a pre-set limit. It enhances safety, ensures legal compliance, and improves fuel efficiency, often required for commercial vehicles

0.5 HP Single-priming Monoblock pump energy- efficient Induction Motor

Dec 2022 - May 2023

Course Project, Under Guidance of Prof. HS Bains (Director)

- Energy-efficient **0.5 HP monoblock pumps** with induction motors are designed for domestic and agricultural water management, offering low power consumption and high durability. These pumps typically feature **copper windings**, thermal overload protection, and are optimized for steady performance even during voltage fluctuations

TECHNICAL SKILLS

- Programming Languages:** Python, MATLAB, Mathematic, Minitab, Cloud Computing
- Tools:** Autodesk Fusion, Fusion 360, SolidWorks, AutoCAD Mechanical, Microsoft Excel

KEY SUBJECT

- Master of Engineering:** Computer Aided Design, Computer Aided Manufacturing, Applied Numerical Method, Kinematics Dynamics and Control
- Bachelor of Engineering:** Strength of Materials, Theory of Machines, Mechanical Vibrations, Design of Mechanical Systems, Manufacturing Processes, Engineering Drawing with CAD

CERTIFICATION

Centre of Excellence - SIEMENS:

- CNC Turning Operation - 5 Days
- Learn Delta- AutoCAD (Isometric Vs 3D Drawing) – 5 Days
- Workshop in Eicher, Chandigarh University - 2 Day